MAP-Net: Multiple Attending Path Neural Network for Building Footprint Extraction From Remote Sensed Imagery

Qing Zhu, Cheng Liao, Han Hu, Xiaoming Mei, and Haifeng Li, *Member, IEEE*

***Abstract*— Building footprint extraction is a basic task in the fields of mapping, image understanding, computer vision, and so on. Accurately and efficiently extracting building footprints from a wide range of remote sensed imagery remains a challenge due to the complex structures, variety of scales, and diverse appearances of buildings. Existing convolutional neural network (CNN)-based building extraction methods are criticized for their inability to detect tiny buildings because the spatial information of CNN feature maps is lost during repeated pooling operations of the CNN. In addition, large buildings still have inaccurate segmentation edges. Moreover, features extracted by a CNN are always partially restricted by the size of the receptive field, and large-scale buildings with low texture are always discontinuous and holey when extracted. To alleviate these problems, multiscale strategies are introduced in the latest research works to extract buildings with different scales. The features with higher resolution generally extracted from shallow layers, which extracted insufficient semantic information for tiny buildings. This article proposes a novel multiple attending path neural network (MAP-Net) for accurately extracting multiscale building footprints and precise boundaries. Unlike existing multiscale feature extraction strategies, MAP-Net learns spatial localization-preserved multiscale features through a multiparallel path in which each stage is gradually generated to extract high-level semantic features with fixed resolution. Then, an attention module adaptively squeezes the channel-wise features extracted from each path for optimized multiscale fusion, and a pyramid spatial pooling module captures global dependence for refining discontinuous building footprints. Experimental results show that our method achieved 0.88%, 0.93%, and 0.45% F1-score and 1.53%, 1.50%, and 0.82% intersection over union (IoU) score improvements without increasing computational complexity compared with the latest HRNetv2 on the Urban 3-D, Deep Globe, and WHU data sets, respectively. Specifically, MAP-Net outperforms multiscale aggregation fully convolutional network (MA-FCN), which is the state-of-the-art (SOTA) algorithms with postprocessing and model voting strategies, on the WHU data set without pretraining and postprocessing. The TensorFlow implementation is available at https://github.com/lehaifeng/MAPNet.**

***Index Terms*— Attention mechanism, building footprint extraction, deep learning, remote sensing imagery, semantic segmentation.**

I. Introduction

THE rapid development of remote sensing technology has made it easier to acquire many high-resolution optical remote sensing images that support the extraction of building footprints in a wide range [1]–[3]. Immediate and accurate building footprint information is significant for illegal building monitoring, 3-D building reconstruction, urban planning, and disaster emergency response. Due to low interclass variance and high intraclass variance in buildings in optical remote sensed imagery [4], parking lots, roads, and other nonbuildings are highly similar to buildings in appearance. With the variety of building materials, scales, and illumination, the representation of buildings in remote sensed imagery shows significant differences. Therefore, how to accurately and efficiently extract building footprints from remote sensed imagery remains a challenge.

Over the past two decades, numerous algorithms have been proposed to extract building footprints. They can be divided into two categories: traditional image processing- and convolutional neural network (CNN)-based methods.

Traditional building extraction methods utilize the characteristics of the spectrum, texture, geometry, and shadow [5]–[10] to design feature operators for extracting buildings from optical images. Since these features vary under different illumination conditions, sensor types, and building architectures, traditional methods can resolve only specific issues for specific data. Awrangjeb *et al.* [11], Du *et al.* [12], Gilani *et al.* [13], and Sohn and Dowman [14] combined optical imagery with GIS data [15], and digital surface models (DSMs) were obtained from light detection and ranging (Lidar) or synthetic aperture radar interferometry [16] to distinguish nonbuilding areas that are highly similar to buildings, which increased the robustness of building extraction, although a wide range of corresponding multisource data acquisition is always costly.

Because buildings in remote sensing images are diverse in structure, appearance, and scale, building extraction algorithms have evolved from handcrafted feature-based methods to learning feature-based methods, such as deep convolutional neural networks (DCNNs). Moreover, deep networks have been practical in designing CNN models. For instance, ResNet [17] with

Manuscript received April 23, 2020; revised June 15, 2020 and August 8, 2020; accepted September 18, 2020. Date of publication October 5, 2020; date of current version June 24, 2021. This work was supported in part by the National Key Research and Development Program of China under Grant 2016YFB0502601 and in part by the National Natural Science Foundation of China under Grant 41631174 and Grant 41871276. *(Corresponding author: Haifeng Li.)*

Qing Zhu, Cheng Liao, and Han Hu are with the Faculty of Geosciences and Environmental Engineering, Southwest Jiaotong University, Chengdu 611756, China (e-mail: zhuq66@263.net; liaocheng@my.swjtu.edu.cn; han.hu@swjtu.edu.cn).

Xiaoming Mei and Haifeng Li are with the School of Geosciences and Info-Physics, Central South University, Changsha 410083, China (e-mail: meixiaoming17@163.com; lihaifeng@csu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2020.3026051

1558-0644 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:04:20 UTC from IEEE Xplore. Restrictions apply.

152 layers introduced identity mapping in a residual block to solve the problem of gradient explosion in propagation, making it possible to design a deeper network to extract richer semantic features.

Evolving from CNN, fully convolutional network (FCN)-based [16], [18]–[27] building extraction methods achieve incredible results and are often used in building semantic segmentation tasks. The encoder–decoder framework [3], [24], [28]–[33] obtains more accurate building footprints than FCN-based methods, particularly on the localization of the boundary since they recover spatial details through skip connections to fuse shallow high-resolution features in the decoder stage. Deep encoding network [34] (DE-Net) introduced lately techniques based on the encoder–decoder network for building extraction and achieved higher performance. Nevertheless, coarse features introduced by shallow layers become the main obstacle for accurate building boundaries. Liu *et al.* [29], Sun *et al.* [35], Ji *et al.* [3], and Li *et al.* [36] used a conditional random field (CRF) for postprocessing to refine the edge of footprints, which achieved great improvements on building boundaries. Multiscale aggregation FCN [37] (MA-FCN) extracted multiscale features based on feature pyramid network and applied polygon regularization methods for boundary refinement that achieved state-of-the-art (SOTA) result on the WHU data set.

For the problem of multiscale building extraction, [25], [29], [38] integrated hierarchical results extracted from multiple models or a design-specific CNN architecture to address multiscale input for accurately extracting multiscale buildings. CU-Net [2] applied multiconstraints based on FCN to enhance multiscale feature representation and performed better in building segmentation than FCN. SiU-Net [39] designed two weight-shared branches for multiscale input and was reported to improve the segmentation accuracy especially for large buildings, yet obviously increased the computational complexity. SRI-Net [3] aggregated multiscale features that generated from modified ResNet101 by a spatial residual inception module, which significantly retaining local details due to the higher resolution features throughout the network. Zhao *et al.* [40] proposed a pyramid spatial pooling module by introducing several global pooling layers to capture multiscale features without significantly increasing the computational complexity. It is more efficient than [25], [29] in multiscale building extraction and improves continuously since the global dependence is captured by pooling layers. EU-Net [4] proposed a deep spatial pyramids pooling (DSPP) module for multiscale feature extraction, which benefits for extracting localization details and multiscale buildings. Besides, it introduced focal loss to reduce the impact of mislabeling promoting the training more stability.

The attention mechanism [41]–[47] is another method for capturing global relations with long-range dependences in spatial or channel aspects, which effectively improves the performance of segmentation. Recently, HRNetv1 [48] and HRNetv2 Sun *et al.* [49] proposed a high-resolution CNN to address multiscale feature extraction and achieved a new goal in semantic segmentation. While the features with different resolutions were fused each other during the feature extraction, the fine boundary details of building localization information may be lost.

In previous studies, CNN-based building footprint extraction algorithms have mainly been encoder–decoder-based, which loses the spatial details in the encoder stage and recovers by fusing shallow feature maps during the decoder stage. However, it causes inaccurate localization on building boundaries since the coarse features introduced from shallow layers and small buildings may be unrecognized. In addition, the extracted features are always partially restricted by the local respective field, and large-scale buildings with low texture are always discontinuous and holey when extracted.

Although many studies have noted the importance of multiscale features for accurate building extraction, there are still many aspects that need to be improved. This research proposes a multiple attending path neural network (MAP-Net) inspired by HRNetv2 [49] to solve the problems described earlier. First, a parallel multipath network is generated gradually in each stage, which consists of serial convolution blocks to extract high-level semantics and preserve multilevel localization details through serial convolution blocks with fixed spatial resolution in each path. Then, the attention mechanism-based feature enhancement module is introduced to adaptively squeeze channel-wise feature maps from each path for multiscale feature optimal combination. Pyramid spatial pooling operations follow to extract global semantic information for continuous building footprints. The main contributions of this study are as follows.

1) We propose an MAP-Net for efficient and accurate multiscale building footprint boundary extraction through parallel localization-preserved convolutional networks. Different from SOTA methods such as HRNetv2, our basic hypothesis is the independence and singularity of path features to be more beneficial. Hence, our strategy is that features extracted from each path were independent with fixed scales, and multiscale features were only fused at the end of the encoder by the attention module.
2) We introduce a channel-wise attention module to adaptively squeeze multiscale features extracted from the multipath network. These features strengthen the building representation by optimally combining global semantic and spatial localization.
3) We validate the effectiveness of the introduced modules in MAP-Net and high-resolution features extracted from shallow layers on building extraction through extensive ablation studies.
4) The proposed method achieved 0.88%, 0.93%, and 0.45% F1-score and 1.53%, 1.50%, and 0.82% intersection over union (IoU) score improvements compared with the latest HRNetv2 [49] on the Urban 3-D [50], Deep Globe [51], and WHU [39] data sets and outperforms other SOTA methods on the WHU data set without pretraining.

The rest of this article is organized as follows. Section II introduces the detailed structure of the proposed network for building extraction. Section III describes the experiments and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:04:20 UTC from IEEE Xplore. Restrictions apply.

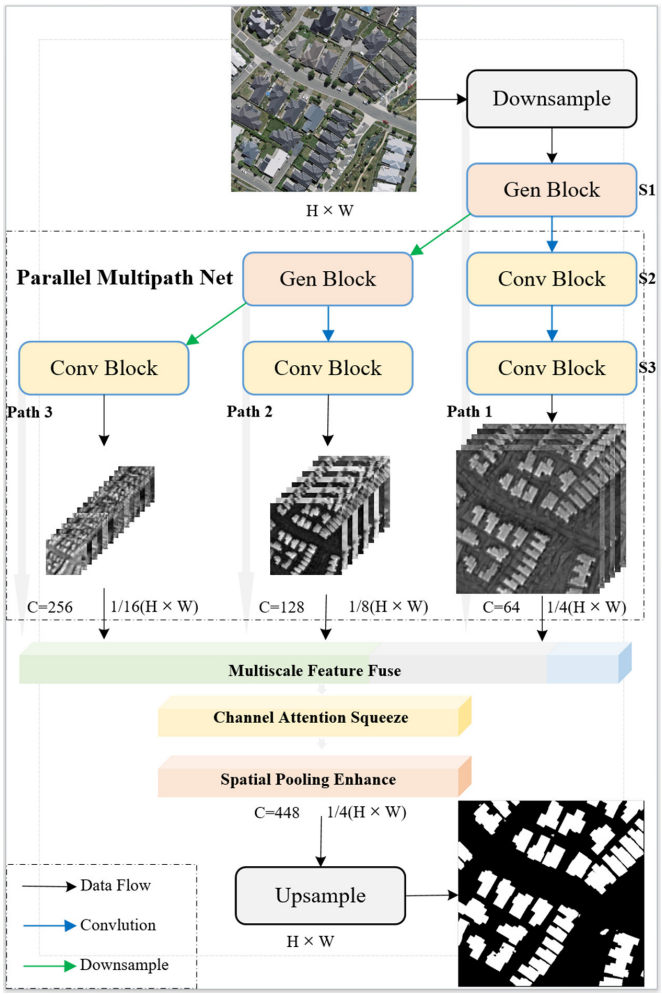


Fig. 1. Structure of the proposed MAP-Net, which is composed of three modules. (a) Detail-preserved multipath feature extraction network. (b) Attention-based multiscale features adaptive squeeze and global spatial pooling enhancement module. (c) Upsampling and building footprint extraction module. The conv block is composed of a series of residual modules to extract features. A gen block generates a new parallel path to extract higher level semantic features based on the conv block.

analyzes the results. The discussions and conclusions of this article are presented in Section IV.

II. Methodology

A. Overview

Repeated pooling layers or stride convolution lose spatial localization during the feature extraction procedure. Existing CNN-based building extraction methods recover spatial localization through skip connections to fuse shallow feature maps or upsample feature maps extracted from the last layer by interpolation. However, shallow feature maps contain coarse semantics, introducing noise information in building extraction. In addition, convolutions process the local neighborhood information hardly to capture global dependence for large buildings. We propose a MAP-Net for multiscale building footprint extraction with accurate boundaries and continuous entities. Fig. 1 shows the structure of the proposed MAP-Net. It mainly includes three components:

1) a parallel multipath network to extract multiscale high-level semantic features while preserving spatial detail information through fixed feature resolution in every path;
2) an attention-based multiscale feature adaptive squeeze and spatial pooling module for global semantic enhancement;
3) an interpolation-based upsampling module for building footprint extraction.

As shown in Fig. 1, the detail-preserved multiscale feature extraction network includes three stages, and the parallel path is generated gradually in each stage to extract richer high-level semantic representations with fixed feature spatial resolution to preserve local details. At the end of the feature extract module, multiscale features extracted from each path were upsampled to the same resolution as features extracted from path 1 by bilinear interpolation and concatenated fusion for attention-based adaptive optimization. The spatial pooling module extracts global dependence to suppress the holes and obtain continuous building entities in the final extraction module. The numbers of channels C and resolution of the feature maps are marked in the figure. H and W represent the height and width of the input image, respectively.

The remainder of this section is arranged as follows. Section II presents a detail-preserved multipath feature extraction network. Attention-based multiscale features adaptively squeeze, and spatial global pooling enhancement is described in Section III. Finally, Section IV describes the basic unit and training strategies involved in this study.

B. Localization-Preserved Multipath Network

Compared with the encoder–decoder-based CNN structure, the advantage of a localization-preserved multipath feature extraction network is that it extracts multiscale features that contain rich high-level semantic representation and accurate spatial localization information rather than recovering spatial details by fusing shallow feature maps during the decoding. Multiscale features extracted from different stages are fed into several parallel paths that are gradually generated to extract richer semantics and preserve spatial resolution without increasing the computational complexity of the network.

Fig. 2 shows part of the proposed multipath network. There are two parallel-path extracted feature maps with different dimensions in the previous stage, and a new path is generated in the next stage to extract higher level semantic features with double-downsampled resolution and double channels as the green arrow shows. It is composed of a max-pooling layer to downsample resolutions and a 1×1 convolutional layer to increase feature channels. The blue arrow represents a 3×3 convolutional layer that transports the features from the last convolution block to the next one. The convolution block was composed of four residual blocks to extract high-level semantic representation, and details were described in Section IV. Feature maps extracted in each path maintain spatial resolution, and richer semantic was extracted as the depth of convolution layers increase. Features with higher resolution preserved as many localization details and lower one captured more global semantic. The most important difference from

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:04:20 UTC from IEEE Xplore. Restrictions apply.

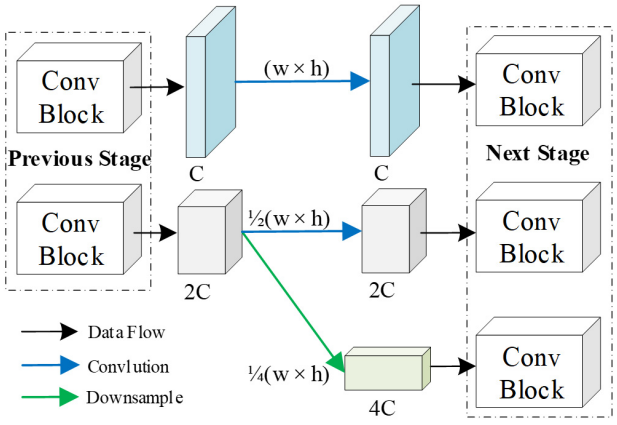


Fig. 2. Part of the detail-preserved multipath network, w and h represent the with and height of the features, respectively. There are two parallel paths with multiscale features extracted in the previous stage. A new path is generated in the next stage to extract higher level semantic representations with downsampled resolution and doubled channels. Features in parallel paths are not fused between two stages.

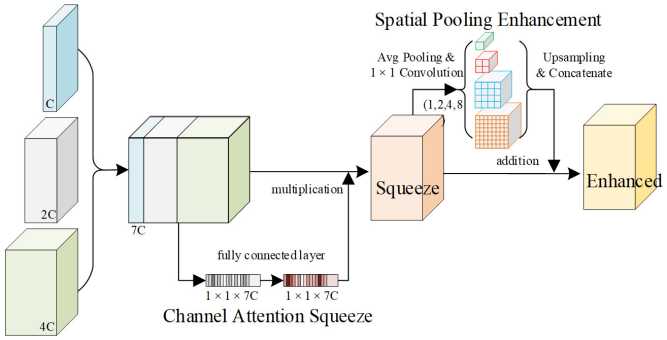


Fig. 3. Feature semantic enhancement module. First, multiscale features extracted from multipaths are scaled by bilinear interpolation and concatenated. Then, the channel attention enhancement module adaptively squeezes significant channel-wise features to reconstruct optimal features. Finally, a spatial pooling enhancement module is introduced to capture global dependence for continuous building footprints.

HRNetv2 is that multiscale features extracted from different paths do not fuse each other during extraction since the fusion operation may weaken localization details.

In the entire process of feature extraction, spatial resolutions and channels of feature maps in each path are fixed. Features in each path are extracted by a series of convolutional blocks that suppress the coarse semantics in high-resolution feature maps compared with encoder–decoder-based CNN. Because the detailed representation is preserved in higher resolution features, smaller buildings and localization of the boundary can be extracted exactly. The effect of a multipath network that considers preserved localization and high-level semantics is explained in experiment 3.C.1).

Considering the tradeoff between complexity and accuracy, MAP-Net is composed of three parallel paths for extracting multiscale features as analyzed in experiment 3.C.2). The resolutions of feature maps are 1/4, 1/8, and 1/16 of the original image, with the corresponding channels that are 64, 128, and 256 in each path.

C. Attention-Based Feature Squeeze

Feature maps extracted from multipaths have different dimensions. Higher resolution features contain localization details and high-level semantic information, whereas lower resolution provides richer global features. The features are upsampled to 1/4 of the original image through bilinear interpolations and fused by concatenation, as shown in Fig. 3. It is important to explore the optimal channel-wise combination of multiscale features since they contain multilevel building localization details. We introduced a channel attention squeeze module that adaptively measures the significance of each channel for optimizing the multiscale features. A spatial pooling enhancement module is introduced to capture global dependence for continuously extracting building entities, especially for large buildings with low texture. The details are described as follows.

In previous CNN-based methods [3], [4], [25], [29], [38], multiscale features were concatenated directly for final pixel-wise prediction. Nevertheless, in our research, multiscale features from different paths contain multilevel spatial localization and richer semantic representation. features with different scales have a dissimilar influence on building extraction. It is the same as each channel, and some of them may weaken the semantic representation but increase the computational complexity. In our research, multiscale features from different paths contain spatial localization and richer semantic representation. It is necessary to distinguish valuable channel-wise features for accurately and efficiently extracting buildings, while *a priori* knowledge hardly weights the importance of each channel. The attention-based feature adaptive squeeze module inspired by [47] plays a role in learning the weight for each channel and automatically reconstructing the feature maps for optimal representation.

As shown in Fig. 4, a global average pooling operation produces a vector of length 7C from the concatenated multiscale channel-wise feature, a fully connected layer with a weight parameter of 7C × 7C, followed by learning a weight vector with a length of 7C corresponding to each channel. The parameters of the fully connected layer are randomly initialized and gradually learned from the features. Finally, the vector that represents the significance of each channel is normalized by a sigmoid function and multiplied to the original features for reconstructing enhanced feature maps.

Due to the extracted features that are always partially restricted by local receptive fields, a spatial pooling module is introduced to extract global dependence. The implementation is similar to [40], except that the global features are generated by four average pooling layers with different sizes that are designed in accordance with the dimensions of features and added to the original feature maps pixel-wise for global spatial enhancement. It captures global relations spatially, which cannot be extracted from the CNN for the local respective field. Hence, extracted buildings have better integrity.

D. Basic Block and Training Strategy

To avoid noise information contained in shallow features due to local receptive fields and decrease the computational complexity, a downsampling block is introduced to decrease

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:04:20 UTC from IEEE Xplore. Restrictions apply.

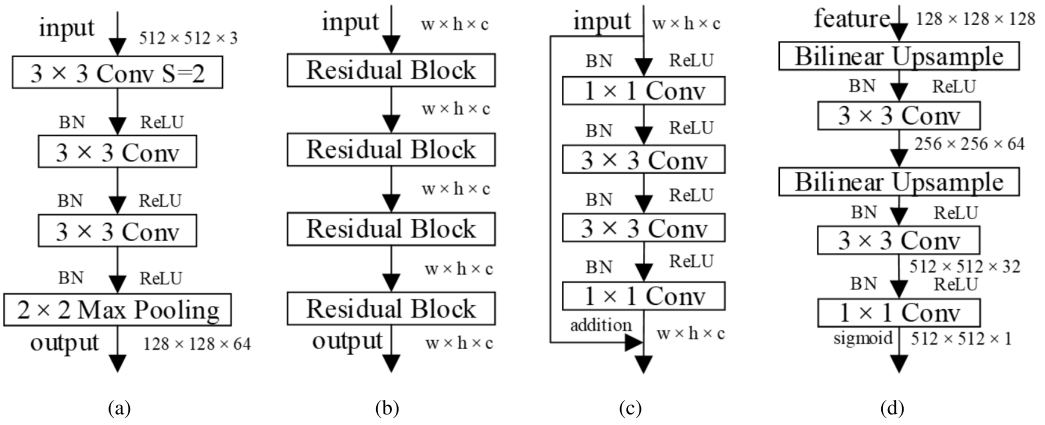


Fig. 4. Detail of the basic blocks. (a) Downsample block. (b) Conv Block. (c) Residual block. (d) Upsample block.

the resolution of the features before the multipath network, as shown in Fig. 4(a). It consists of a stride convolution layer, two 3×3 convolutional layers, and a max-pooling layer to extract feature maps with 64 channels and 1/4 spatial resolution of the input image. Fig. 4(b) shows the conv block, which includes several residual blocks in series. The impact of different numbers of blocks on performance is explored in experiment 3.3.2. The residual block consists of a 1×1 convolutional layer for reducing the dimensions of features, two 3×3 layers for extracting features, and a 1×1 convolutional layer for restoring dimensions to the input; a shortcut fuses input to output through element-wise addition, and batch normalization (BN) and rectified linear unit (ReLU) execute before the convolutional layers, as shown in Fig. 4(c). The building footprint extraction module is shown in Fig. 4(d), the resolution of the features is recovered through bilinear interpolation in two stages, and the convolutional layers are used to decrease the number of channels. The output layer is a single-channel feature map with the same spatial resolution as input, and each value represents the probability that it belongs to building.

Our research was implemented in TensorFlow using a single 2080Ti GPU with 12-GB memory. The Adam optimizer was chosen with an initial learning rate of 0.001, and beta1 and beta2 were set to default as recommended. All compared methods were trained from scratch for approximately 80 epochs until convergence and randomly rotated and flipped for data augmentation on three building data sets described in Section III. The batch size was set to 4, given that the restrictions of the GPU memory size and the same hyperparameters were maintained to compare the performance with different methods for equality.

Sigmiod cross-entropy loss was selected as the loss function because of the pixel-wise binary classification involved. The computation of loss at position (i, j) is given as (1); logits represent the predicted result and $y_{ij} \in \{0, 1\}$ represents the ground truth; the sigmoid function was applied to logits to ensure that $p \in [0, 1]$, as shown in (2). The loss value is the average of Loss_{ij} at all positions for an input image

$$\text{Loss}_{ij} = -[y_{ij} * \ln p_{ij} + (1 - y_{ij}) \ln(1 - p_{ij})] \quad (1)$$

$$p_{ij} = \text{Sigmoid}(\text{logits}_{ij}) = \frac{1}{1 + e^{-\text{logits}_{ij}}}. \quad (2)$$

III. Experiment and Analysis

A. Data Set

To evaluate the proposed method, we conducted a comprehensive experiment on three open data sets, including the WHU building data set [39], the Deep Globe Building Extraction Challenge data set [51], and the Urban 3-D Challenge data set [50]. The details are described as follows.

The WHU building data set includes both aerial and satellite subsets with the corresponding shapefiles and raster images. In our experiment, we selected the aerial subset, which has various appearances and scales of buildings, to evaluate the robustness of the proposed algorithm. It consists of more than 187 000 buildings, covering over a 450 km^2 area, with 30-cm ground resolution. Each image has three bands, corresponding to red (R), green (G), and blue (B) wavelengths, with each image size of 512×512 pixels. There is a total of 8188 tiles of images, including 4736, 2416, and 1036 tiles as training, test, and validation data sets, respectively. We conducted our experiment at its original provided data set partitioning.

The Deep Globe Building Extraction Challenge data set [51] contains WorldView-3 satellite imagery captured from Las Vegas, Paris, Shanghai, and Khartoum. In this research, the Las Vegas and Shanghai subsets were selected to evaluate the generalization performance of the proposed algorithm. There were approximately 243 382 buildings with 30-cm ground resolution, covering over 1216 km^2, and the size of each image was 650×650 pixels. All images were randomly divided as 6:1:3 as the training, validation, and test sets, respectively.

The Urban 3-D Challenge data set [50] contains 208 orthorectified RGB, with corresponding DSM and digital terrain models (DTMs) generated from commercial satellite imagery. It contains approximately 157 000 buildings, covering over 360 km^2 with a ground resolution of 50 cm, and the size of each image is 2048×2048 pixels. DSM and DTM indicate the elevation of buildings, which obviously improves the building extraction performance. We used only the RGB images in our experiment to evaluate the performance of the proposed method. The training, validation, and test sets include 104, 62, and 42 tiles, respectively, as the original data partition

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:04:20 UTC from IEEE Xplore. Restrictions apply.

method, and we randomly clipped the images to the size of 512×512 pixels for training and testing.

B. Evaluation Metric

Generally, evaluation metric methodologies can be divided into two categories: pixel-level metrics and instance-level metrics. The pixel-level method counts the correctly classified and misclassified pixels pixel-wise. In the instance-level method, a building is correctly extracted only when the IoU between the prediction and ground truth is larger than a specific threshold. Semantic segmentation-based building footprint extraction aims to classify every pixel, whether or not it belongs to a building, for a specific input image. Therefore, we apply a pixel-level metric, including precision, recall, F1-score, and IoU, to evaluate the performance of MAP-Net and other different methods.

There are four classifying conditions: true prediction on a positive sample (TP), false prediction on a positive sample (FP), true prediction on a negative sample (TN), and false prediction on a negative sample (FN). Precision represents the percentage of TP in total positive prediction, recall indicates the percentage of TP over the total positive samples, the F1-score is the weighted average of precision and recall, which considers both FP and FN, and IoU is the average value of the intersection of the prediction and ground truth over their union of the whole image set. Equations are given as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{3}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{4}$$

$$\text{F1} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

$$\text{IoU} = \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall} - \text{Precision} * \text{Recall}}. \tag{6}$$

C. Experimental Setup

In this section, we first analyzed the significance of the proposed multipath architecture for extracting multiscale buildings with exact localization on boundaries compared with the popular encoder–decoder framework. Second, we explored the impact of different network parameters on the complexity and accuracy of MAP-Net on a specific data set. Third, a contrast experiment was carried out to compare the performance of MAP-Net with four classic semantic segmentation algorithms on building extraction. Then, we compared the performance of most recent studies on building extraction based on the WHU data set. Finally, we conducted an ablation experiment to validate the significance of the proposed network and analyzed the tradeoff between complexity and accuracy among the compared methodologies. Details are described in the following sections.

1) Significance of Multipath: Feature maps extracted from the proposed localization-preserved multipath network are shown in Fig. 5. Columns (b)–(d) are extracted from the path (P) 1 with the same spatial resolution (R) on each end of the stage (S) corresponding to the sample image in column (a). This indicates that feature maps with higher resolution extracted from deeper convolutional layers (larger S) retained richer semantics; in other words, building and background could be distinguished evidently. Columns (d)–(f) show the extracted feature maps from each path at the end of stage 3 with decreasing spatial resolution, as shown in Fig. 1. It shows that feature maps with lower resolution are more blurred at the edge of buildings, and in worse conditions, small buildings may be lost completely, as shown in column (f), due to the exact localization lost in downsampling operation during the feature extraction procedure.

Encoder–decoder-based networks fuse higher resolution feature maps extracted from shallow layers, such as columns (b) or (c), to recover exact localization through a skip connection at the decoder stage, which introduces noise information for the coarse semantic features. In addition, small buildings may be lost in the lowest resolution feature maps, such as column (f), which cannot be refined accurately during the decoder stage. As a result, extracted building footprints were inaccurate on the boundary, or worse, small buildings were unrecognized.

Multipath networks extract multiscale feature maps through parallel paths. The resolutions of feature maps in each path were fixed, multiscale features from the different paths were not fused in the whole feature extraction process. Features with higher spatial resolution preserved exact localization and contained rich semantic information through a deeper CNN, such as column (d). It is beneficial for extracting fine boundaries and small buildings compared to skip connection with coarse shallow features as well as multiscale features fusion such as HRNetv2. In addition, the features with lower spatial resolution captured global semantic representations, which contribute to the extraction of large buildings. Multiscale features extracted from multipaths were combined and enhanced at the end of stage 3 to extract buildings with multiscales, which makes up for the shortcoming of the existing network.

2) Network Parameter: The structure of the proposed network is mainly affected by the depth of the convolution network and the number of parallel paths on a specific data set. Without loss of generality, we designed an experiment to explore the potential impact of different structures on the performance of MAP-Net on the WHU data set. Readers should be noticed that the network parameters will be different as data set changes.

The depth is represented by the number of residual blocks (N-blocks) in each path; empirically, these were set from 3 to 6 in our experiments that consider the tradeoff between accuracy and complexity. Similarly, the number of paths (N-paths) was chosen from 2 to 4 according to the resolution of the input image. The IoU metric was used to evaluate the performance, and the number of trainable parameters (Para.) was counted to represent the complexity of the network.

The experimental result is shown in Fig. 6 with the increase in N-blocks, and the IoU score increased first and then decreased after N-blocks were greater than a specific value, which may be explained by the complexity of the network growing with N-blocks increasing while weakening the generalization ability of the model. However, the Para.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:04:20 UTC from IEEE Xplore. Restrictions apply.

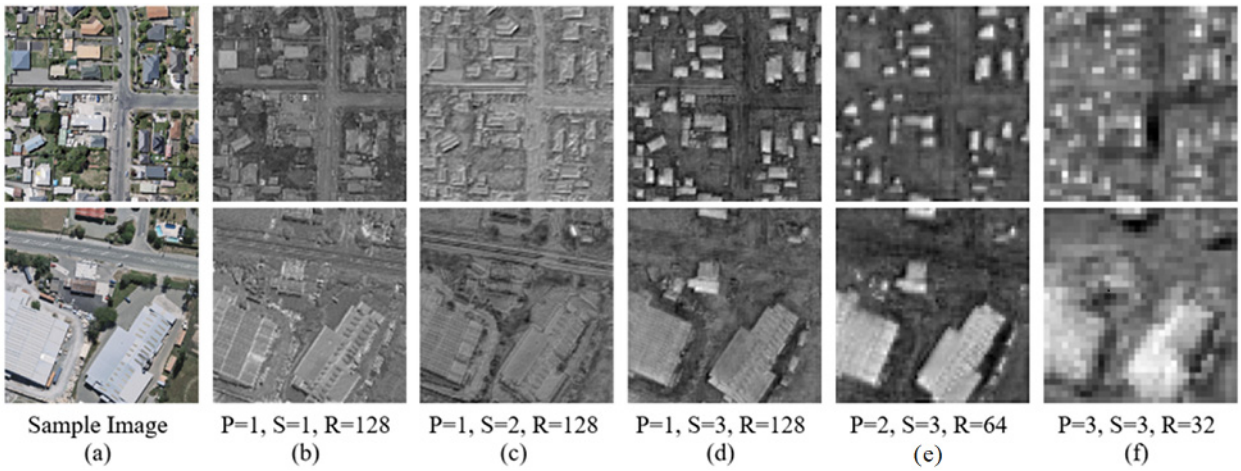


Fig. 5. Feature maps extracted from localization-preserved multipath networks with different paths (P), stages (S), and spatial resolutions (R) referred to in Fig. 1. (a) Two sample images containing multiscale buildings. (b)–(d) Feature maps extracted from path 1 with the same spatial resolution at the end of each stage. (d)–(f) Feature maps extracted from stage 3 but with decreasing spatial resolution in each path.

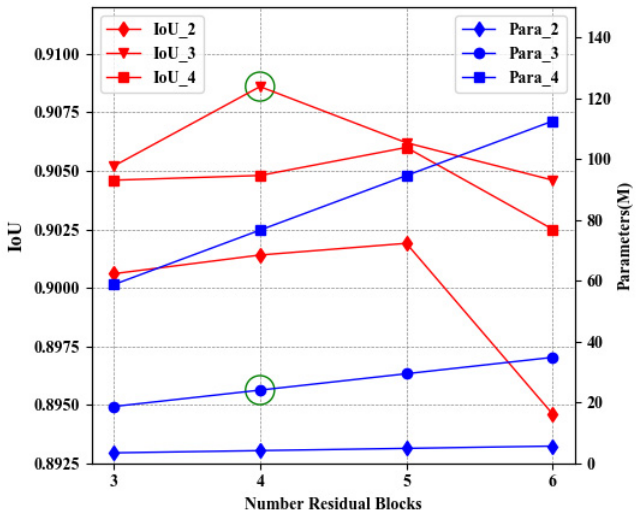


Fig. 6. Performance of MAP-Net with different network structures. The diamond, circle, and rectangle represent different numbers of paths. Red and blue represent the IoU score and trainable parameters (Para.), respectively. The horizontal axis indicates the number of residual blocks (N-blocks) in a convolutional block.

grows linearly with the increase in N-blocks while increasing exponentially with the increase in N-paths since the generated path doubles the feature channels, which greatly increases the parameters during the feature extraction and enhancement stage.

Features with specific resolution were extracted from each path. The different number of paths impacts the combination of multiscale semantic features fused in MAP-Net. When the N-paths were equal to 3, the IoU metric was better than that of 2 or 4, as shown by the solid line marked by the red circle in Fig. 6.

Considering the balance between accuracy and complexity, the better structure of the MAP-Net was composed of three parallel paths, and each convolutional block consisted of four residual blocks, which contained fewer parameters and performed better than others on the WHU data set, as indicated by solid circles marked with green.

Besides, channels (C) of feature maps in each path and resolution of input images may also affect the performance of the proposed method. Generally, larger C will lead to better performance in the case of enough training data but increase complexity exponentially. Optimal paths of MAP-Net are probably affected by the resolution of remote sensing imagery for the different combinations of multiscale features representation. The better structure of MAP-Net on another data set could be explored similarly. As it is not the focus of this article, we will not state the detailed discussion here.

3) Performance Evaluation: To evaluate the performance of the proposed network, we conducted contrast experiments to compare MAP-Net with four SOTA methods, including UNet, PSPNet with ReNet50 backbone, ResNet101, and HRNetv2, on data sets [39], [50], [51].

U-NetPlus achieved great improvement through a redesigned encoder by VGG11 and replaced the transposed convolution with nearest-interpolation based on U-Net, which widely used in remote sensing imagery segmentation. Since ResNet improved the training stability and performance of deeper CNN by introducing residual connection, we reimplemented PSPNet with ResNet50 for feature extraction and modified ResNet101 for building extraction through replacing the upsample module with the same as MAP-Net.

Experimental results are shown in Tables I–III. Our proposed method demonstrates a great improvement compared with other methods on three experimental data sets and obtains approximately 0.82%, 1.50%, and 1.53% IoU improvement and 0.45%, 0.93%, and 0.88% F1-score improvement on the WHU data set, Deep Globe data set, and Urban 3-D data set, respectively, compared with the latest research HRNetv2. The best records are marked with bold.

To compare different methods, some example results on each data set are presented in Figs. 7–9. Fig. 7 shows the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:04:20 UTC from IEEE Xplore. Restrictions apply.

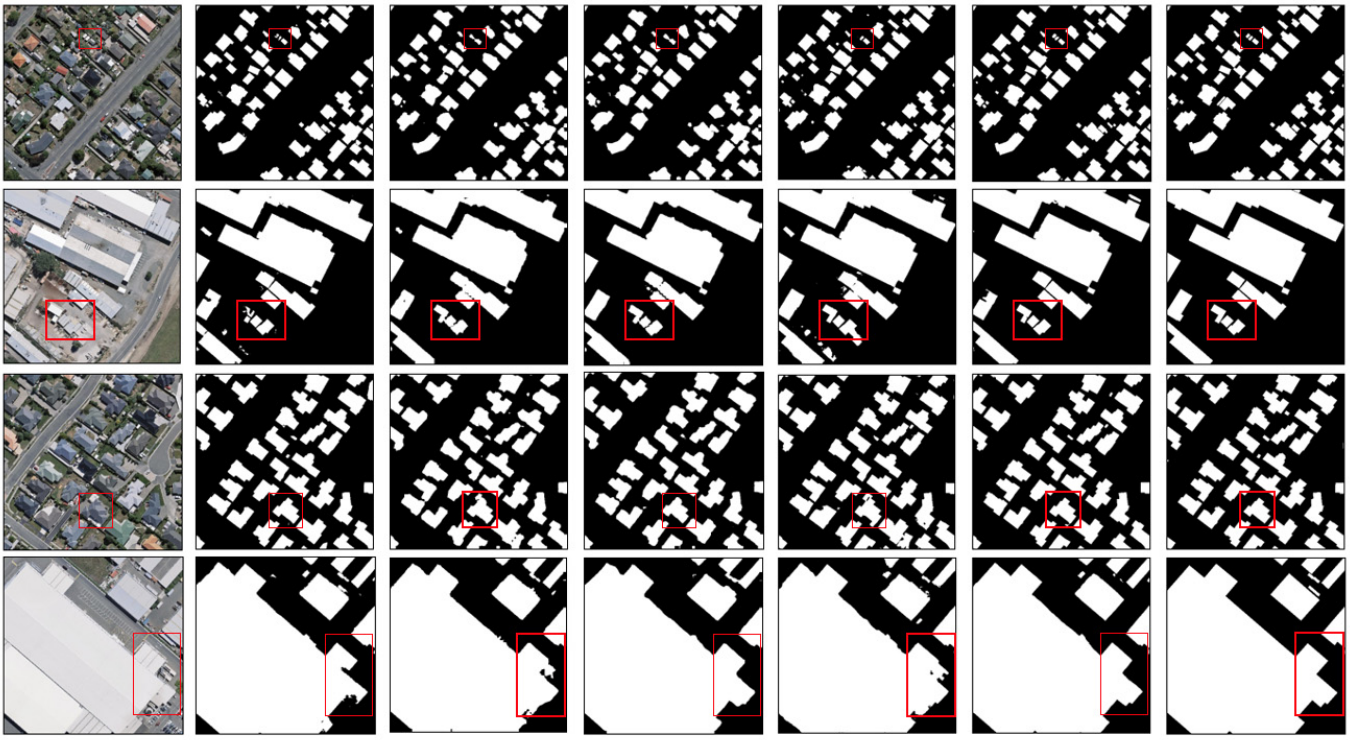


Fig. 7. Example of results with the U-NetPlus, PSPNet, ResNet101, HRNetv2, and our proposed method on the WHU data set. (a) Original image. (b) U-NetPlus. (c) PSPNet. (d) ResNet101. (e) HRNetv2. (f) Ours. (g) Ground truth.

TABLE I
COMPARISON OF THE STATE-OF-THE-ART METHODS AND OURS ON THE WHU DATA SET

Method	IoU(%)	Precision(%)	Recall(%)	F1-score(%)
U-NetPlus	88.75	94.85	93.25	94.04
PSPNet	88.87	94.28	93.93	94.10
ResNet101	89.18	94.47	94.09	94.28
HRNetv2	90.04	94.16	**95.37**	94.76
Ours	**90.86**	**95.62**	94.81	**95.21**

TABLE II
COMPARISON OF THE STATE-OF-THE-ART METHODS AND OURS ON THE DEEP GLOBE DATA SET

Method	IoU(%)	Precision(%)	Recall(%)	F1-score(%)
U-NetPlus	76.34	89.07	84.23	86.59
PSPNet	78.76	87.36	**88.89**	88.12
ResNet101	79.16	89.10	87.65	88.37
HRNetv2	79.13	89.55	87.17	88.35
Ours	**80.63**	**91.29**	87.35	**89.28**

TABLE III
COMPARISON OF THE STATE-OF-THE-ART METHODS AND OURS ON THE URBAN 3-D DATA SET

Method	IoU(%)	Precision(%)	Recall(%)	F1-score(%)
U-NetPlus	84.56	92.59	90.69	91.63
PSPNet	86.19	93.00	92.17	92.46
ResNet101	86.17	92.83	92.31	92.57
HRNetv2	86.15	92.74	92.38	92.56
Ours	**87.68**	**93.42**	**93.45**	**93.44**

extracted building footprints on the WHU data set. There are four examples, including buildings with various appearances and scales. Columns (a) and (g) represent the original image and corresponding ground truth, and columns (b)–(f) are extracted results from U-NetPlus, PSPNet, ResNet101, HRNetv2, and MAP-Net, respectively.

The results show that our proposed method outperforms the other four compared methods obviously, especially by more accurately recognizing small buildings and more completely extracting large buildings, which benefits from the localization-preserved multipath feature extraction network and the multiscale feature adaptively enhancement module. The boundary of buildings is more exactly based on the ground truth.

Example results on the Deep Globe data set and Urban 3-D data set are shown in Figs. 8 and 9. Each column has the same meaning, as shown in Fig. 7.

4) Comparison of Recent Methods: We compare our methods to the most recent building extraction methods, including CU-Net [2], SiU-Net [39], SRI-Net [3], DE-Net [34],

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:04:20 UTC from IEEE Xplore. Restrictions apply.

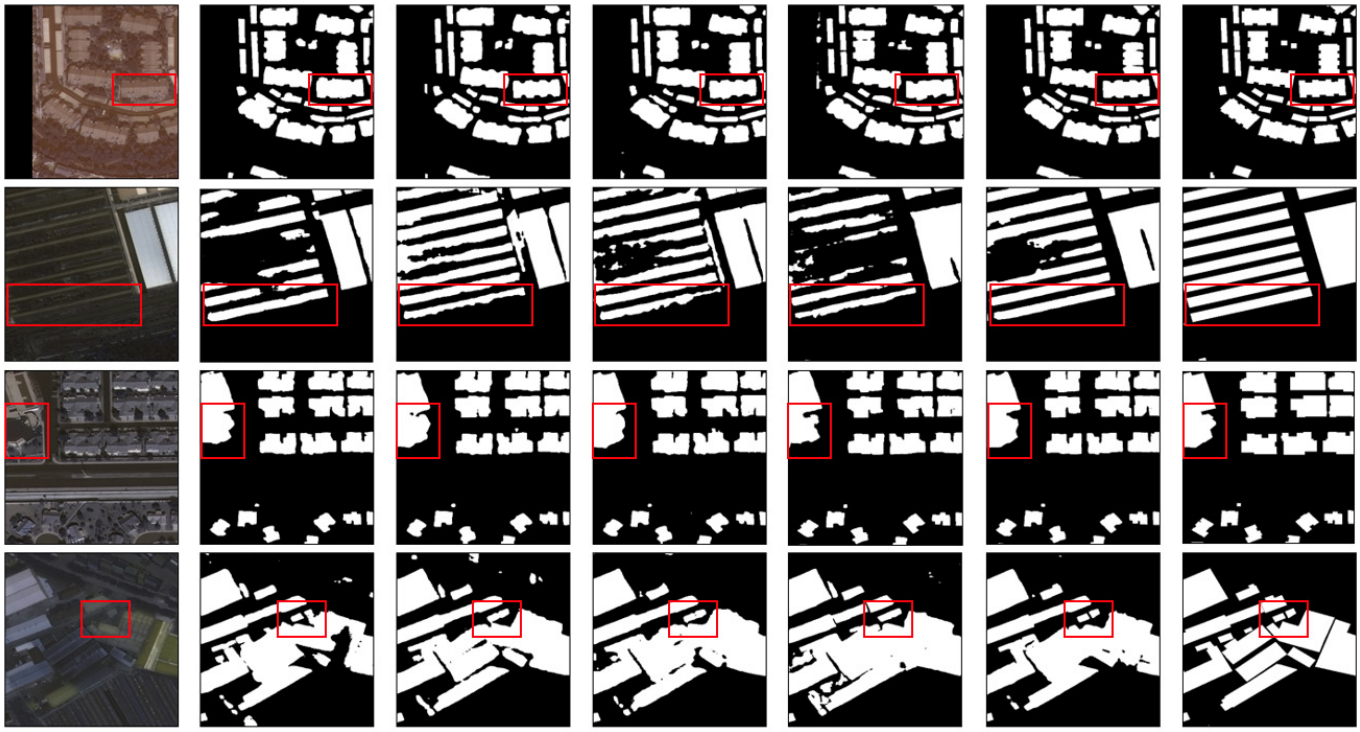


Fig. 8. Example of results with the U-NetPlus, PSPNet, ResNet101, HRNetv2, and our proposed method on the Deep Globe data set. (a) Original image. (b) U-NetPlus. (c) PSPNet. (d) ResNet101. (e) HRNetv2. (f) Ours. (g) Ground truth.

TABLE IV
COMPARISON OF MOST RESENT BUILDING EXTRACTION METHODS ON THE WHU DATA SET

Method	IoU(%)	Precision(%)	Recall(%)	F1-score(%)
CU-Net [2]	87.10	94.60	91.70	93.13
SiU-Net [39]	88.40	93.80	93.90	93.85
SRI-Net [3]	89.23	**95.67**	93.69	94.51
DE-Net [34]	90.12±0.24	95.00±0.16	94.60±0.19	94.80±0.18
EU-Net [4]	90.56	94.98	**95.10**	95.04
MA-FCN [37]	90.70	95.20	**95.10**	95.15
MAP-Net (Ours)	**90.86**	95.62	94.81	**95.21**

EU-Net [34], and MA-FCN [37]on the WHU test data set to evaluate the performance of MAP-Net. Table IV shows the segmentation result of recent studies and ours, and the best records are marked with bold. We did not reproduce their results since the source codes are not available.

CU-Net [2] introduced multiconstraint to enhance feature representation based on FCN. SiU-Net [39] designed multiscale input with two weight-shared UNets that improve the result, especially for a large building. DE-Net [34] introduced lately segmentation techniques for building extraction based on the encoder–decoder network and achieved IoU accuracy higher than 90%. Considering the significance of multiscale features for building extraction, SRI-Net [3] generated multilevel features through modified ResNet101 backbone and aggregate multiscale contexts by spatial residual inception module, which achieved 89.23% IoU on the WHU data set. EU-Net [34] outperformed previous methods by a large margin with 90.56% IoU since it captures multiscale features by designed a DSPP and introduced focal loss to reduce the impact of mislabeling, which benefit for extracting fine details and multiscale buildings. Most recent MA-FCN extract multiscale features based on feature pyramid network considering the multiscale building extraction and many postprocessing strategies for boundary refinement. It is better than EU-Net and achieved SOTA results, which significantly benefits from multimodel voting strategy on the WHU data set.

As shown in Table IV, our study achieved 90.86% in IoU metric and 95.21% in F1-Score metric that outperform all the most recently studies. Especially, it slightly outperforms MA-FCN on a result with limited performance improvement space without any pretraining and postprocessing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:04:20 UTC from IEEE Xplore. Restrictions apply.

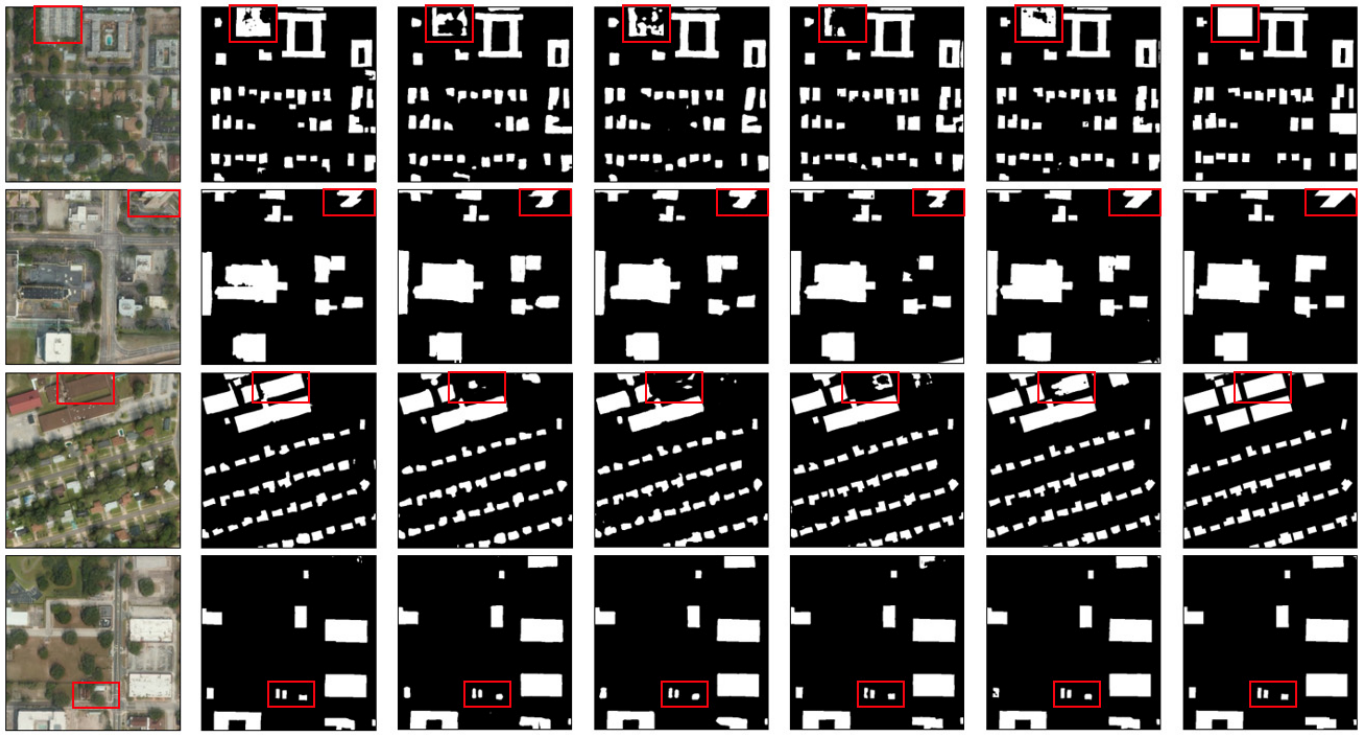


Fig. 9. Example of results with the U-NetPlus, PSPNet, ResNet101, HRNetv2, and our proposed method on the Urban 3-D data set. (a) Original image. (b) U-NetPlus. (c) PSPNet. (d) ResNet101. (e) HRNetv2. (f) Ours. (g) Ground truth.

TABLE V
INFLUENCE OF DIFFERENT MODULES. (M): MULTIPATH EXTRACTION. (C): CHANNEL FEATURE SQUEEZE. (S): SPATIAL FEATURE ENHANCE. (F) CONNECTION WITH SHALLOW FEATURES

No.	Method	IoU(%)	Precision(%)	Recall(%)	F1-Score (%)
1	HRNetv1	89.54	94.55	94.41	94.48
2	HRNetv2	90.04	94.16	**95.37**	94.76
3	Baseline	90.28	95.03	94.76	94.89
4	Baseline+ C	90.57	95.44	94.67	95.05
5	Baseline+ S	90.62	95.21	94.95	95.08
6	MAP-Net(Ours)	90.86	**95.62**	94.81	95.21
7	MAP-Net + F	**90.90**	95.18	95.29	**95.23**

As shown in Table IV, our studies proposed a localization-preserved multipath feature extraction network with a feature enhancement module for building extraction, which achieved 90.86 IoU and outperformed the latest MA-FCN without pretraining and postprocessing on the WHU data set.

5) Ablation Experiments: To explore the contributions of different modules of the MAP-Net, we conducted ablation experiments on the WHU data set and evaluate the accuracy of IoU, precision, recall, and F1-score.

First, we validated the performance of the localization-preserved strategy (Baseline) compared with HRNetv2 and HRNetv1. Second, we explore the effect of feature squeeze module and global enhancement module on MAP-Net. Finally, we compare the influence of shallow features (F) that have a half resolution of the input imagery on MAP-Net with skip connection.

Experimental results are recorded in Table V, the highest values are highlighted in bold, and the visualization of extracted results from each method is shown in Fig. 10 for the comparison of different methods. The false prediction marked with red, the false-positive marked with blue, and the true positive marked with green.

Our baseline based on HRNetv2 optimized the architecture on multiscale features fusion as described in Section II-B, and the baseline methods outperform the HRNetv2 by 0.24% on the IoU since modified multipath localization-preserved strategy without feature fusion in the encoder and higher

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:04:20 UTC from IEEE Xplore. Restrictions apply.

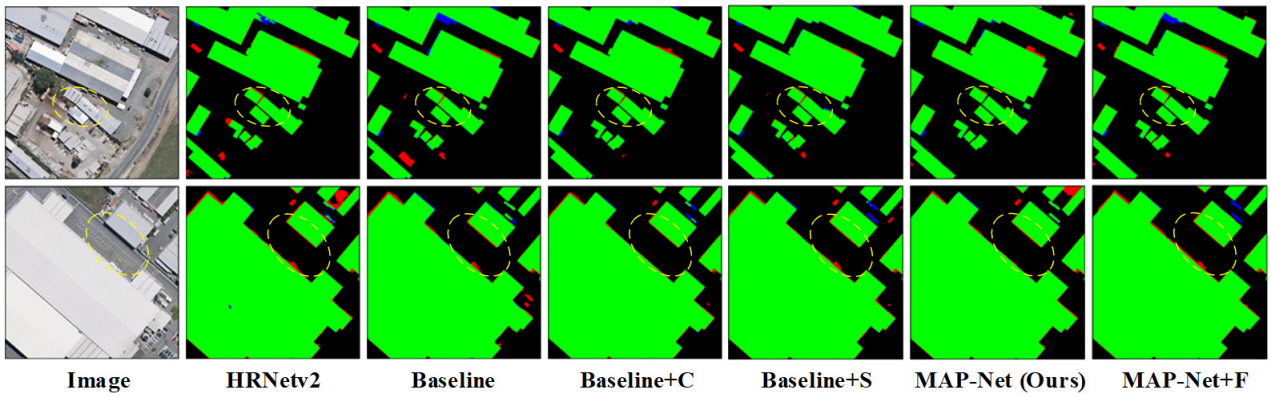


Fig. 10. Visualization of extracted results on ablation experiments. False prediction marked with red, false positive marked with blue, and true positive marked with green.

0.74% than the HRNetv1 on the IoU for fusing multiscale features on the decoder the same as HRNetv2.

Based on the baseline, the channel-wise feature adaptively squeezes module based on the attention mechanism called (C) and the spatial pooling enhancement module named (S) improved the IoU performance by 0.29% and 0.34%, respectively. The MAP-Net achieved 0.58% IoU arise with feature squeeze module and global enhancement module compared to our baseline. As described in Fig. 1, the resolution of fused features is a quarter of input image in our MAP-Net. To evaluate the influence of shallow feature that has a higher resolution on building extraction, we concatenate the feature with half-resolution of input image extracted from the downsample block to the upsample block thought skip connection named MAP-Net + F.

According to the experimental results, adding shallow features improved the accuracy by 0.04%, but the introduced shallow features also increase the coarse noise information, as shown in the last column of Fig. 10, resulting in inaccurate building boundary. The MAP-Net can get a smoother edge of the building without introducing shallow features and reduce unnecessary error recognition with little accuracy loss.

It is worth noting that our algorithms from No. 3 to 6 obtained higher precision measures than HRNetv2 with the same threshold equals to 0.5. A probable explanation is that our methods suppressed false-positive prediction, which contributed to accurate multiscale features extracted from localization-preserved multipath networks, without fusion during feature extraction. The same conclusion can be inferred from other data sets according to Tables II and III.

6) Complexity of MAP-Net: Our proposed algorithm extracted features with multiscales; specifically, some paths needed to process feature maps with large resolution to preserve exact localization in the whole network, which could lead to large numbers of parameters. To validate the trade-off between the performance and complexity of MAP-Net, we compared FLOPs, trainable parameter, and IoU score of related methods on the WHU data set.

As shown in Table VI, the U-NetPlus has the lowest complexity but with poor performance. ResNet101 is the most complicated model due to the most numbers of convolutional layers and the highest numbers of channels among the related methods. HRNetv2 has slightly more parameters than HRNetv1 and performs better since the concatenate fusion of multiscale features in the decoder stage.

The complexity of our baseline has reduced with higher performance compared to HRNetv2 since the redesign structure of encoder. Although the MAP-Net has increased complexity

TABLE VI
COMPARISON OF RELATED METHODS ON FLOPS, TRAINABLE PARAMETERS, AND IOU SCORE ON THE WHU TEST DATA SET

Method	FLOPs(M)	Parameters(M)	IoU(%)
U-NetPlus	17.28	8.636	89.51
PSPNet	93.48	46.73	88.87
ResNet101	104.31	52.11	89.18
HRNetv1	57.18	28.54	89.54
HRNetv2	59.20	29.54	90.04
Baseline	46.78	23.35	90.28
Baseline + C	47.16	23.54	90.63
Baseline + S	47.58	23.75	90.56
MAP-Net (Ours)	48.09	24.00	90.86

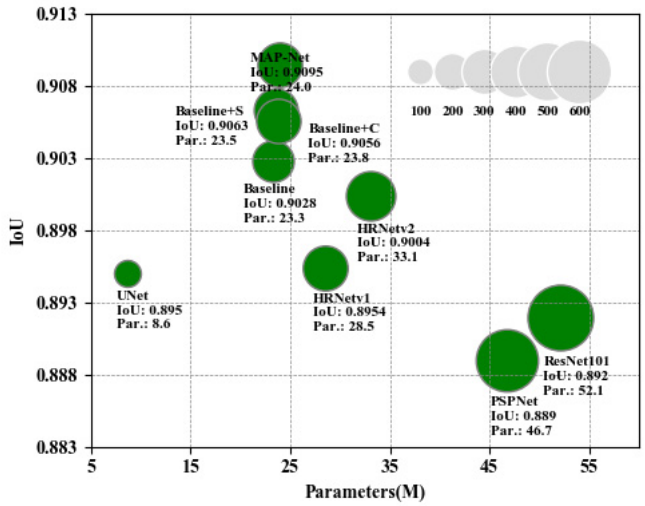


Fig. 11. Complexity and accuracy comparison among related methods. IoU precision and the number of trainable parameters for each method are marked in the figures. The radius of the circle indicates the size of the model file.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:04:20 UTC from IEEE Xplore. Restrictions apply.

after introducing a feature enhancement module, the performance has greatly improved with far fewer FLOPs than HRNetv2.

Although MAP-Net maintains a high-resolution feature map in the whole feature extraction process, which may lead to a large number of parameters, the number of channels remains small, allowing it to efficiently extract multiscale features. To compare the complexity and performance among related methods more intuitively, we presented the experimental results in Fig. 11. The number of trainable parameters and IoU precision represents complexity and performance for each related method. The radius of the green circle indicates the size of the model file. Our proposed methods maintain higher accuracy and lower complexity compared with other related methods.

IV. Discussion and Conclusion

To solve the problem of extracted building footprints with inaccurate boundaries and possibly lost small buildings as well as discontinuous for large-scale buildings, in this research, we proposed a novel localization-preserved multipath feature extraction network inspired by HRNetv2 with an adaptively multiscale feature optimal fusion module and spatial enhancement module for building footprint extraction. Multiscale features extracted from parallel multipaths that contain multilevel local details, as well as rich semantic representations allow it to accurately extract building footprints with exact edges and recognize small buildings. The enhanced module further reconstructs and optimizes features in channel and spatial aspects, which suppresses the holes and extracts continuous footprints for large buildings.

The experiments on three different benchmarks demonstrate that the MAP-Net outperforms other classic semantic segmentation algorithms with higher accuracy and lower complexity and achieved SOTA result on the WHU data set among the most recent building extraction methods. In addition, we conducted an ablation experiment to evaluate the significance of the proposed module and proved that localization-preserved multipath network extraction of buildings achieves higher precision than previous methods.

Generally, our research provides a new approach for accurately and efficiently extracting multiscale objects that are common in the real world. Currently, our experiments are implemented in building extraction, and we will further study multiclass extraction tasks, such as land cover, to achieve automatic interpretation of remotely sensed imagery in future work.

References

[1] K. Bittner, F. Adam, S. Cui, M. Korner, and P. Reinartz, "Building footprint extraction from VHR remote sensing images combined with normalized DSMs using fused fully convolutional networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 8, pp. 2615–2629, Aug. 2018.

[2] G. Wu *et al.*, "Automatic building segmentation of aerial imagery using multi-constraint fully convolutional networks," *Remote Sens.*, vol. 10, no. 3, p. 407, Mar. 2018.

[3] P. Liu *et al.*, "Building footprint extraction from high-resolution images via spatial residual inception convolutional neural network," *Remote Sens.*, vol. 11, no. 7, p. 830, Apr. 2019.

[4] W. Kang, Y. Xiang, F. Wang, and H. You, "EU-net: An efficient fully convolutional network for building extraction from optical remote sensing images," *Remote Sens.*, vol. 11, no. 23, p. 2813, Nov. 2019.

[5] J. Du *et al.*, "A novel framework for 2.5-D building contouring from large-scale residential scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 4121–4145, Jun. 2019.

[6] Z. Li, W. Shi, Q. Wang, and Z. Miao, "Extracting man-made objects from high spatial resolution remote sensing images via fast level set evolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 2, pp. 883–899, Feb. 2015.

[7] N. L. Gavankar and S. K. Ghosh, "Automatic building footprint extraction from high-resolution satellite image using mathematical morphology," *Eur. J. Remote Sens.*, vol. 51, no. 1, pp. 182–193, 2018.

[8] J.-P. Burochin, B. Vallet, M. Brédif, C. Mallet, T. Brosset, and N. Paparoditis, "Detecting blind building façades from highly overlapping wide angle aerial imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 96, pp. 193–209, Oct. 2014.

[9] M. Cote and P. Saeedi, "Automatic rooftop extraction in nadir aerial imagery of suburban regions using corners and variational level set evolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 1, pp. 313–328, Jan. 2013.

[10] G. Zhou and X. Zhou, "Seamless fusion of LiDAR and aerial imagery for building extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 7393–7407, Nov. 2014.

[11] M. Awrangjeb, C. Zhang, and C. S. Fraser, "Automatic extraction of building roofs using LIDAR data and multispectral imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 83, pp. 1–18, Sep. 2013.

[12] S. Du, Y. Zhang, Z. Zou, S. Xu, X. He, and S. Chen, "Automatic building extraction from LiDAR data fusion of point and grid-based features," *ISPRS J. Photogramm. Remote Sens.*, vol. 130, pp. 294–307, Aug. 2017.

[13] S. Gilani, M. Awrangjeb, and G. Lu, "An automatic building extraction and regularisation technique using LiDAR point cloud data and orthoimage," *Remote Sens.*, vol. 8, no. 3, p. 258, Mar. 2016.

[14] G. Sohn and I. Dowman, "Data fusion of high-resolution satellite imagery and LiDAR data for automatic building extraction," *ISPRS J. Photogramm. Remote Sens.*, vol. 62, no. 1, pp. 43–63, May 2007.

[15] E. Simonetto, H. Oriot, and R. Garello, "Rectangular building extraction from stereoscopic airborne radar images," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 10, pp. 2386–2395, Oct. 2005.

[16] X. X. Zhu *et al.*, "Deep learning in remote sensing: A comprehensive review and list of resources," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 8–36, 2017.

[17] K. He, X. Zhang, S. Ren, and J. Sun, "Identity mappings in deep residual networks," in *Proc. Eur. Conf. Comput. Vis.* Springer, 2016, pp. 630–645.

[18] Z. Huang, G. Cheng, H. Wang, H. Li, L. Shi, and C. Pan, "Building extraction from multi-source remote sensing images via deep deconvolution neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 1835–1838.

[19] R. Kemker, C. Salvaggio, and C. Kanan, "Algorithms for semantic segmentation of multispectral remote sensing imagery using deep learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 60–77, Nov. 2018.

[20] Y. Sun, X. Zhang, Q. Xin, and J. Huang, "Developing a multi-filter convolutional neural network for semantic segmentation using high-resolution aerial imagery and LiDAR data," *ISPRS J. Photogramm. Remote Sens.*, vol. 143, pp. 3–14, Sep. 2018.

[21] E. Maltezos, A. Doulamis, N. Doulamis, and C. Ioannidis, "Building extraction from LiDAR data applying deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 1, pp. 155–159, Jan. 2018.

[22] Z. Deng, H. Sun, S. Zhou, J. Zhao, L. Lei, and H. Zou, "Multi-scale object detection in remote sensing imagery with convolutional neural networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 3–22, Nov. 2018.

[23] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "Convolutional neural networks for large-scale remote-sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 645–657, Feb. 2017.

[24] K. Bittner, S. Cui, and P. Reinartz, "Building extraction from remote sensing data using fully convolutional networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.-ISPRS Arch.*, vol. 42, pp. 481–486, Jan. 2017.

[25] G. Sun, H. Huang, A. Zhang, F. Li, H. Zhao, and H. Fu, "Fusion of multiscale convolutional neural networks for building extraction in very high-resolution images," *Remote Sens.*, vol. 11, no. 3, p. 227, Jan. 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:04:20 UTC from IEEE Xplore. Restrictions apply.

[26] R. Alshehhi, P. R. Marpu, W. L. Woon, and M. D. Mura, "Simultaneous extraction of roads and buildings in remote sensing imagery with convolutional neural networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 130, pp. 139–149, Aug. 2017.
[27] R. Zhang, G. Li, M. Li, and L. Wang, "Fusion of images and point clouds for the semantic segmentation of large-scale 3D scenes based on deep learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 143, pp. 85–96, Sep. 2018.
[28] W. Li, C. He, J. Fang, J. Zheng, H. Fu, and L. Yu, "Semantic segmentation-based building footprint extraction using very high-resolution satellite images and multi-source GIS data," *Remote Sens.*, vol. 11, no. 4, p. 403, Feb. 2019.
[29] S. Ji, S. Wei, and M. Lu, "A scale robust convolutional neural network for automatic building extraction from aerial and satellite imagery," *Int. J. Remote Sens.*, vol. 40, no. 9, pp. 3308–3322, May 2019.
[30] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Springer, 2015, pp. 234–241.
[31] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.
[32] A. Khalel and M. El-Saban, "Automatic pixelwise object labeling for aerial imagery using stacked U-Nets," 2018, *arXiv:1803.04953*. [Online]. Available: http://arxiv.org/abs/1803.04953
[33] S. M. Kamrul Hasan and C. A. Linte, "U-NetPlus: A modified encoder-decoder U-Net architecture for semantic and instance segmentation of surgical instruments from laparoscopic images," in *Proc. 41st Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2019, pp. 7205–7211.
[34] H. Liu *et al.*, "DE-net: Deep encoding network for building extraction from high-resolution remote sensing imagery," *Remote Sens.*, vol. 11, no. 20, p. 2380, Oct. 2019.
[35] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.
[36] J. Lin, W. Jing, H. Song, and G. Chen, "ESFNet: Efficient network for building extraction from high-resolution aerial images," *IEEE Access*, vol. 7, pp. 54285–54294, 2019.
[37] S. Wei, S. Ji, and M. Lu, "Toward automatic building footprint delineation from aerial images using cnn and regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 2178–2189, Mar. 2020.
[38] L. Li, J. Liang, M. Weng, and H. Zhu, "A multiple-feature reuse network to extract buildings from remote sensing imagery," *Remote Sens.*, vol. 10, no. 9, p. 1350, Sep. 2018.
[39] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.
[40] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.
[41] J. He, Z. Deng, L. Zhou, Y. Wang, and Y. Qiao, "Adaptive pyramid context network for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7519–7528.
[42] J. Fu *et al.*, "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.
[43] Y. Yuan and J. Wang, "OCNet: Object context network for scene parsing," 2018, *arXiv:1809.00916*. [Online]. Available: http://arxiv.org/abs/1809.00916
[44] Y. Cao, J. Xu, S. Lin, F. Wei, and H. Hu, "GCNet: Non-local networks meet squeeze-excitation networks and beyond," 2019, *arXiv:1904.11492*. [Online]. Available: http://arxiv.org/abs/1904.11492
[45] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.
[46] Y. Chen, Y. Kalantidis, J. Li, S. Yan, and J. Feng, "A^2-Nets: Double attention networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, pp. 352–361.
[47] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.
[48] K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5693–5703.
[49] K. Sun *et al.*, "High-resolution representations for labeling pixels and regions," 2019, *arXiv:1904.04514*. [Online]. Available: http://arxiv.org/abs/1904.04514
[50] H. R. Goldberg, S. Wang, G. A. Christie, and M. Z. Brown, "Urban 3D challenge: Building footprint detection using orthorectified imagery and digital surface models from commercial satellites," *Proc. SPIE*, vol. 10645, Apr. 2018, Art. no. 1064503.
[51] I. Demir *et al.*, "DeepGlobe 2018: A challenge to parse the Earth through satellite images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 172–17209.



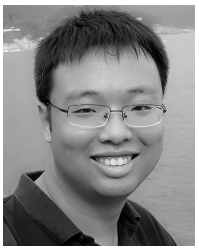
Qing Zhu received the Ph.D. degree from Northern Jiaotong University, Beijing, China.

He was a Postdoctoral Fellow with Wuhan University, Wuhan, China, from 1995 to 1997. He was a Professor with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, from 1998 to 2013. He is a Professor with the Faculty of Geosciences and Environmental Engineering, Southwest Jiaotong University, Chengdu, China. His research interests include the digital elevation model, 3-D GIS, virtual geographic environment, and smart city.



Cheng Liao received the B.Sc. degree in computer science and technology from the School of Information Science and Engineering, Central South University, Changsha, China, in 2017. He is pursuing the M.Sc. degree with the Faculty of Geosciences and Environmental Engineering, Southwest Jiaotong University, Chengdu, China.

His research interests include remote sensing and deep learning.



Han Hu received the B.Sc. degree from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2010, and the Ph.D. degree from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, in 2015.

He was a Postdoctoral Fellow with The Hong Kong Polytechnic University, Hong Kong, from 2015 to 2019. He is a Professor with the Faculty of Geosciences and Environmental Engineering, Southwest Jiaotong University, Chengdu, China.



Xiaoming Mei received the B.S. degree in applied geophysics and the M.S. degree in geodetection and information technology from Central South University, Changsha, China, in 2000 and 2003, respectively, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2007.

His research interests include multispectral and hyperspectral remote sensing data processing, high-resolution image processing and scene analysis, LiDAR data processing, and computational intelligence.



Haifeng Li (Member, IEEE) received the master's degree in transportation engineering from the South China University of Technology, Guangzhou, China, in 2005, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2009.

He was a Research Associate with the Department of Land Surveying and Geo-Informatics, The Hong Kong Polytechnic University, Hong Kong, in 2011, and a Visiting Scholar with the University of Illinois at Urbana–Champaign, Urbana, IL, USA, from 2013 to 2014. He is a Professor with the School of Geosciences and Info-Physics, Central South University, Changsha, China. His research interests include geo/remote sensing big data, machine/deep learning, and artificial/brain-inspired intelligence.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 10:04:20 UTC from IEEE Xplore. Restrictions apply.